Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 1 of 7 · Question 4 part c.i · 2 marks

2023 JUN · Paper 01 · extract 1 of 4 · find part c.i in this mark scheme extract

4(a)	**The only correct answer is C** ***A*** *is not correct because* ***mebibits*** *is a measure of storage capacity* ***B*** *is not correct because* ***mebibytes*** *is a measure of storage capacity* ***D*** *is not correct because transmission speeds are measured only in bits per second, not* ***bytes per second***		1

Question Number	Answer	Additional Guidance	Mark
4(b)	Award **one** mark for each correct cell: **Type** / **Example** / **Number of bits** IPv4 / 192.169.0.3 / 32 (1) IPv6 / 1050:a500:00c0:0440:0006:0300:700d:436f / 128 (1)		2

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 1 of 7 · Question 4 part c.i · 2 marks

2023 JUN · Paper 01 · extract 2 of 4 · find part c.i in this mark scheme extract

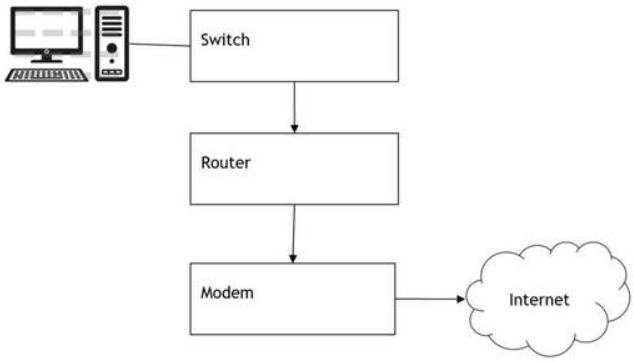
Question Number	Answer	Additional Guidance	Mark
4(c)(i)	Award up to **two** marks for a linked description, such as: • In a wired network packets travel over (copper/fibre-optic) cables (1), whereas in a wireless network packets travel over the air (1)	Allow radio signals, infra-red, microwave for air (must be more than just waves) Allow twisted pair, CAT for copper. Allow electrical/light signals (not just electricity)	2

Question Number	Answer	Additional Guidance	Mark
4(c)(ii)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. A protocol (suite/group/family) (1) 2. Used on a wired network / wired connection (1) 3. Defines physical parts/type of cable(twisted pair, CAT6)/type of connector (1) 4. Defines how packets are checked for errors (1) 5. Defines the speed of transmission (1) 6. Operates at the link layer of the TCP/IP stack (1)		2

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 1 of 7 · Question 4 part c.i · 2 marks

2023 JUN · Paper 01 · extract 3 of 4 · find part c.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(d)	Award **three** marks for all of switch, router and modem in the correct order as in diagram Award **two** marks for all of switch, router and modem in any order Award **one** mark for any of switch or router or modem in any box 		3

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 1 of 7 · Question 4 part c.i · 2 marks

2023 JUN · Paper 01 · extract 4 of 4 · find part c.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(e)	Award up to **two** marks for a linked description, such as: Validation is used to check that the data/password entered by a user is fit for processing/meets specified requirements (1), whereas authentication is used to verify the identity of the user/prove that users are who they say they are (1)	For validation 'specified requirements' a good description of the check can be given e.g. presence check, length check, format check	2

Question Number	Answer	Additional Guidance	Mark

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 2 of 7 · Question 3 part b · 1 mark

2019 JUN · Paper 01 · extract 1 of 4 · find part b in this mark scheme extract

3(a)(i)	• LAN / Local area network (1) Allow PAN/Personal Area Network		1

Question Number	Answer	Additional Guidance	Mark
3(a)(ii)	• WAN / Wide area network (1)		1

Question Number	Answer	Additional Guidance	Mark
3(b)	A: Megabits per second **The only correct answer is A** **B** is not correct because megapixels are associated with images rather than network data speeds **C** is not correct because mebicycles are not associated with network data speeds **D** is not correct because mebihertz are not associated with network data speeds		1

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 2 of 7 · Question 3 part b · 1 mark

2019 JUN · Paper 01 · extract 2 of 4 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(c)	• IMAP (1) AND **one** from: • because changes are synced (1) with the mail server in real-time (1) • because the messages always reside on the email server (1) as she has limited storage (1) on her tablet and phone • it gets sent to both smartphone and tablet (1) **Alternative answer:** • POP3 (not awarded a mark) AND **one** from: • because storage limitations on server (1) and privacy issues (1) • because more CPU time (1) may be used when syncing (1) (a large number of messages).		3

Question Number	Answer	Additional Guidance	Mark
3(d)	<table><tr><th>Characteristic</th><th>Bus</th><th>Ring</th><th>Star</th><th>Mesh</th></tr><tr><td>Network performance degrades as more devices are added.</td><td>✓</td><td></td><td></td><td></td></tr><tr><td>All devices are directly or indirectly connected to all others.</td><td></td><td></td><td></td><td>✓</td></tr><tr><td>Each node has a physical attachment to a routing device.</td><td></td><td></td><td>✓</td><td></td></tr><tr><td>There are no data collisions, because packets travel in the same direction.</td><td></td><td>✓</td><td></td><td></td></tr></table>		4

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 2 of 7 · Question 3 part b · 1 mark

2019 JUN · Paper 01 · extract 3 of 4 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(e)(i)	-81 One mark for negative sign One mark for 81.		2

Question Number	Answer	Additional Guidance	Mark
3(e)(ii)	One mark for each nibble in correct order 0100 1011		2

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 2 of 7 · Question 3 part b · 1 mark

2019 JUN · Paper 01 · extract 4 of 4 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(e)(iii)	Any **two** from: 1. This is an overflow (1) error 2. It is caused by adding two 1s in the most significant bit (1) 3. The resulting number is too large to fit in the number of bits available / requires 9 bits / the number is larger than 2^8 / there is 1 that needs to be carried over (in MSB) / MSB gets discarded (1) 4. It can cause the program to crash or produce unreliable/incorrect results (1)		2

Question Number	Answer	Additional Guidance	Mark

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 3 of 7 · Question 1 part b.i · 2 marks

2020 NOV · Paper 01 · extract 1 of 5 · find part b.i in this mark scheme extract

1(a)	Any **two** from: • (Share) access to the Internet/WWW/broadband connection (1) • (Internal) communication (using email/instant messaging/calendars) (1) • Share files/data (1) • Share peripherals/printers / hardware (1) • Increases the amount of storage (since network storage devices can be attached) (1) • Saves money on licences (since network site licences for software are usually cheaper than buying a stand-alone licence for every machine) (1) • Centralised backup (1) • Centralised security (1)	• Award examples if mapped to a bullet point • Do not award the same bullet more than once • Do not award for install updates (in the stem)	2

Question Number	Answer	Additional Guidance	Mark
1(b) (i)	Any **two** from: • ~~Faster (data transfer) (1)~~ Not really true, its Greater bandwidth / more bits per second (1) • The connection does not get worse the further you are from the router / more reliable (1) • Connection does not get obstructed by walls, ceilings, and furniture (1) • More secure (1)	Accept reverse arguments for disadvantages of wireless.	2

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 3 of 7 · Question 1 part b.i · 2 marks

2020 NOV · Paper 01 · extract 2 of 5 · find part b.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(b) (ii)	Any **two** from: • Installation/maintenance is more complex (1) • Devices need to be physically connected (1) • Less portable / limited by length of cable (1) • Limited number of devices can be connected (1) • Some digital devices cannot use a wired connection (1) • Trip hazard (1)	Accept reverse arguments for advantages of wireless.	2

Question Number	Answer	Additional Guidance	Mark
1(c)	Award **two** marks for a linked description that addresses the individual mark points: The internet is a (global) network of networks/ connected devices (1) whereas the www is the collection of web pages/service accessed using the internet (1) / the www is resources located via URLs/domain names (1) The internet is the infrastructure (1) and the www is a service that runs on that infrastructure (1)		2

Question Number	Answer	Additional Guidance	Mark

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Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 3 of 7 · Question 1 part b.i · 2 marks

2020 NOV · Paper 01 · extract 3 of 5 · find part b.i in this mark scheme extract

1(d)	Award **two** marks for a linked explanation such as: Running out of IPv4 addresses/the number of (internet) connected devices has grown (1) IPv6 addresses are long/longer than IPv4 / the number of possible addresses is large/will last much longer (1)		2

Question Number	Answer	Additional Guidance	Mark
1(e)(i)	Star		1

Question Number	Answer	Additional Guidance	Mark
1(e) (ii)	Any **one** from: • easy to connect/remove new nodes (1) • failure of one node/link does not affect the rest of the network (1) • easy to detect the failure of one node/link (1)		1

Question Number	Answer	Additional Guidance	Mark
1(e) (iii)	Any **one** from: • if central switch/hub fails then the whole network fails (1) • performance and number of nodes that can be added depends on capacity of central switch/hub (1) • can require more cable than some of the other topologies (1)		1

Question Number	Answer	Additional Guidance	Mark

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 3 of 7 · Question 1 part b.i · 2 marks

2020 NOV · Paper 01 · extract 4 of 5 · find part b.i in this mark scheme extract

1(f)(i)	Any **two** from: • SMTP (1) • IMAP (1) • POP/POP3 (1)		2

Question Number	Answer	Additional Guidance	Mark
1(f)(ii)	Any **two** from: • passes the (reassembled) packets to the application layer (1) • check if all packets have arrived (1) • determine whether the contents are correct (1) • requests resending of lost or damaged packets (1) • reassembles packets in correct order/ into an email (1)	Do not accept receives from network layer (stem)	2

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Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 3 of 7 · Question 1 part b.i · 2 marks

2020 NOV · Paper 01 · extract 5 of 5 · find part b.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 4 of 7 · Question 1 part b.ii · 2 marks

2020 NOV · Paper 01 · extract 1 of 5 · find part b.ii in this mark scheme extract

1(a)	Any **two** from: • (Share) access to the Internet/WWW/broadband connection (1) • (Internal) communication (using email/instant messaging/calendars) (1) • Share files/data (1) • Share peripherals/printers / hardware (1) • Increases the amount of storage (since network storage devices can be attached) (1) • Saves money on licences (since network site licences for software are usually cheaper than buying a stand-alone licence for every machine) (1) • Centralised backup (1) • Centralised security (1)	• Award examples if mapped to a bullet point • Do not award the same bullet more than once • Do not award for install updates (in the stem)	2

Question Number	Answer	Additional Guidance	Mark
1(b) (i)	Any **two** from: • ~~Faster (data transfer) (1)~~ Not really true, its Greater bandwidth / more bits per second (1) • The connection does not get worse the further you are from the router / more reliable (1) • Connection does not get obstructed by walls, ceilings, and furniture (1) • More secure (1)	Accept reverse arguments for disadvantages of wireless.	2

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 4 of 7 · Question 1 part b.ii · 2 marks

2020 NOV · Paper 01 · extract 2 of 5 · find part b.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(b) (ii)	Any **two** from: • Installation/maintenance is more complex (1) • Devices need to be physically connected (1) • Less portable / limited by length of cable (1) • Limited number of devices can be connected (1) • Some digital devices cannot use a wired connection (1) • Trip hazard (1)	Accept reverse arguments for advantages of wireless.	2

Question Number	Answer	Additional Guidance	Mark
1(c)	Award **two** marks for a linked description that addresses the individual mark points: The internet is a (global) network of networks/ connected devices (1) whereas the www is the collection of web pages/service accessed using the internet (1) / the www is resources located via URLs/domain names (1) The internet is the infrastructure (1) and the www is a service that runs on that infrastructure (1)		2

Question Number	Answer	Additional Guidance	Mark

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1(d)	Award **two** marks for a linked explanation such as: Running out of IPv4 addresses/the number of (internet) connected devices has grown (1) IPv6 addresses are long/longer than IPv4 / the number of possible addresses is large/will last much longer (1)		2

Question Number	Answer	Additional Guidance	Mark
1(e)(i)	Star		1

Question Number	Answer	Additional Guidance	Mark
1(e) (ii)	Any **one** from: • easy to connect/remove new nodes (1) • failure of one node/link does not affect the rest of the network (1) • easy to detect the failure of one node/link (1)		1

Question Number	Answer	Additional Guidance	Mark
1(e) (iii)	Any **one** from: • if central switch/hub fails then the whole network fails (1) • performance and number of nodes that can be added depends on capacity of central switch/hub (1) • can require more cable than some of the other topologies (1)		1

Question Number	Answer	Additional Guidance	Mark

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 4 of 7 · Question 1 part b.ii · 2 marks

2020 NOV · Paper 01 · extract 4 of 5 · find part b.ii in this mark scheme extract

1(f)(i)	Any **two** from: • SMTP (1) • IMAP (1) • POP/POP3 (1)		2

Question Number	Answer	Additional Guidance	Mark
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Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 4 of 7 · Question 1 part b.ii · 2 marks

2020 NOV · Paper 01 · extract 5 of 5 · find part b.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 5 of 7 · Question 3 part c · 1 mark

2021 NOV · Paper 01 · extract 1 of 5 · find part c in this mark scheme extract

3(a)	Award **one** mark for correct network type Award **one** mark for correct usage model One mark for each column		2

Network type	Tick (□)	Usage model	Tick (□)
Local Area Network (LAN)		Client-Server	
Wide Area Network (WAN)		Peer-to-peer	□
Personal Area Network (PAN)	□		

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 5 of 7 · Question 3 part c · 1 mark

2021 NOV · Paper 01 · extract 2 of 5 · find part c in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>3(b)</td><td>Award one mark for each correct layer name up to a maximum of two
Award one mark per layer for a correct function associated with the correct layer up to a maximum of two

<table>
<tr><th>Layer</th><th>Function</th></tr>
<tr><td>Application</td><td>• Selects and uses the correct protocol to transmit data
• Interacts with the user</td></tr>
<tr><td>Transport</td><td>• Identifies the server port to use
• Identifies the client port to use
• Divides the data into packets
• Numbers the packets
• Adds the total number of packets
• Sets up communication between hosts / Establishes end to end communication
• Passes the packets to the network layer
• Checks the packets arrive at the destination
• Resends any packets that have not arrived</td></tr>
<tr><td>Network</td><td>• Adds the source/sender's IP address
• Adds the destination IP address
• Routes the packets</td></tr>
<tr><td>Data link</td><td>• Controls physical connections between pieces of hardware
• Adds MAC addresses to the packets
• Sends the packets on their way
• Adds headers and trailers</td></tr>
</table></td><td>Accept functions at the receiving end too</td><td>4</td></tr>
</table>

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 5 of 7 · Question 3 part c · 1 mark

2021 NOV · Paper 01 · extract 3 of 5 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(c)	**The only correct answer is A** B *is not correct because* it is not a smartphone frequency band C *is not correct because* it is not a smartphone frequency band D *is not correct because* it is not a smartphone frequency band		1

Question Number	Answer	Additional Guidance	Mark
3(d)(i)	Award **one** mark from: • higher frequency has more waves per second/carries more data in the same time (1) • higher frequency gives greater bandwidth (1) • higher frequency is more stable/less prone to interference (1)		1
3(d)(ii)	Award up to **two** marks for a linked explanation such as: • gives Carlo faster responses to his communications/connections (1) allowing Carlo to move/download/share data in a shorter time (1) • gives Carlo more reliable communications/can access communications more widely/in more places (1) giving Carlo a better user experience (1) • gives Carlo more secure communications (1) improving his privacy/reducing other people's ability to intercept/spy on his communications (1)		2

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 5 of 7 · Question 3 part c · 1 mark

2021 NOV · Paper 01 · extract 4 of 5 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(e)(i)	Award **one** mark from: • do not leave the smartphone on standby/turn off when not being used (1) • use (portable) solar cells/other renewable generation method (1) • unplug charger when not in use (1) • reduce screen brightness (1) • turn off WiFi/GPS/Bluetooth/location (when not in use) (1) • close apps that are not being used (1)		1

Question Number	Answer	Additional Guidance	Mark
3(e)(ii)	Award up to **two** marks for a linked explanation. Award **one** mark for the action • keep smartphone for longer (1) • repair/recycle smartphone if not working (1) • upgrade the operating system (1) Award **one** mark for the impact • less e-waste is generated (1) • manufacturing could be reduced (1)		2
		Total	**13**

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 5 of 7 · Question 3 part c · 1 mark

2021 NOV · Paper 01 · extract 5 of 5 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 6 of 7 · Question 6 part b.ii · 2 marks

2022 JUN · Paper 01 · extract 1 of 4 · find part b.ii in this mark scheme extract

6(a)(i)	Award up to **two** marks for a linked explanation such as: • Can transfer data quickly / reduced chance of packet collisions (1) as data only flows in one direction (1) • No need for a server/switch/hub (1) because the packets do not have to be directly routed to a specific device / each workstation controls connectivity / packets are passed from workstation to workstation until the destination is reached (1) • Every workstation gets equal access to resources (1) because each station has to wait until it gets a token / devices do not have to compete to get a token (1) • Additional workstations can be easily added/easy to set up (1) because each workstation only connects to two other workstations (1) • Easy to find faults (1) all of the tokens will end up on one workstation (1) • Cheap to set up (1) uses minimum cabling (1)	Not a **Comparison** without expansion that fits mark scheme	2
6(a)(ii)	Award up to **two** marks for a linked explanation such as: • It needs to be easily scalable (1) mesh topology allows this as it is decentralised / nodes connect with other nodes around them (1) • Can handle high volumes of data traffic (1), because data can travel via multiple routes (1) • Is self-healing/resilient/allows alternative paths (1), which means data will still reach its destination even if a node or connection fails(1) Not system/computer/switch etc • Enables it to span a huge geographic area (1), because additional nodes can be added to expand coverage (1)		2

Question Number	Answer	Additional Guidance	Mark
6(b)(i)	Award **one** mark for: • Personal area network / PAN/WPAN		1

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 6 of 7 · Question 6 part b.ii · 2 marks

2022 JUN · Paper 01 · extract 2 of 4 · find part b.ii in this mark scheme extract

6(b)(ii)	Award up to **two** marks for a linked explanation such as: • A faster connection speed (1) because fewer users/devices sharing the bandwidth/connection (1) • Improved security /stated security issue(1) because it uses secure cellular data connection / not on public network /Santiago has to approve users(1)		2

Question Number	Answer	Additional Guidance	Mark
6(c)(i)	Award **one** mark from: • A record of activities/specified activity that have taken place on a computer system (1) • Automatic record of what has happened and who did it (1)		1
6(c)(ii)	Award **one** mark from: • To identify suspicious/malicious activity/changes (1) • To increase accountability (1) • To trace a problem back to its source/perpetrator (1) • To find out if any users are using unauthorised applications (capable of putting the network at risk) (1)		1

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 6 of 7 · Question 6 part b.ii · 2 marks

2022 JUN · Paper 01 · extract 3 of 4 · find part b.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
6(d)	**Ethical hacking** • Ethical hackers are white hat hackers • Attempt to access the network as a hacker does • Don't attempt to change or steal data • Looking for weaknesses in the network • Weakness pointed out • Weaknesses fixed • Could be employed by the business • Could work for another specialist company • Can include penetration testing **Commercial analysis tools** • Software used to find weaknesses • Can be configured to check for a range of weaknesses • Results/reports generated identifying faults • Weaknesses fixed **Review of network and user policies** • Collection of rules and guidelines that govern the behaviours of network devices/users • Need reviewing because may not comply with new laws and regulations • Reviews should be scheduled		6
		Total for question 6	**15**

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1–2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured.

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 6 of 7 · Question 6 part b.ii · 2 marks

2022 JUN · Paper 01 · extract 4 of 4 · find part b.ii in this mark scheme extract

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Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 7 of 7 · Question 4 part a · 1 mark

2023 JUN · Paper 01 · extract 1 of 4 · find part a in this mark scheme extract

4(a)	**The only correct answer is C** ***A*** *is not correct because* ***mebibits*** *is a measure of storage capacity* ***B*** *is not correct because* ***mebibytes*** *is a measure of storage capacity* ***D*** *is not correct because transmission speeds are measured only in bits per second, not* ***bytes per second***		1

Question Number	Answer	Additional Guidance	Mark
4(b)	Award **one** mark for each correct cell: **Type** / **Example** / **Number of bits** IPv4 / 192.169.0.3 / 32 (1) IPv6 / 1050:a500:00c0:0440:0006:0300:700d:436f / 128 (1)		2

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 7 of 7 · Question 4 part a · 1 mark

2023 JUN · Paper 01 · extract 2 of 4 · find part a in this mark scheme extract

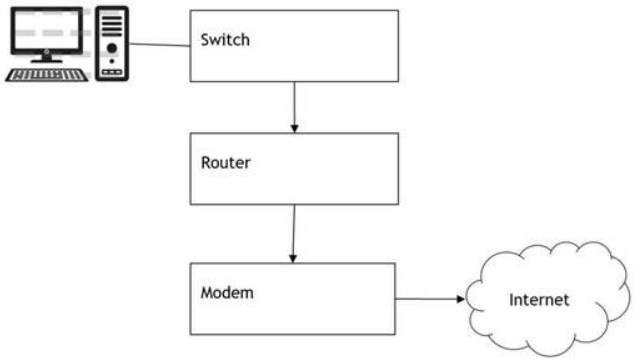
Question Number	Answer	Additional Guidance	Mark
4(c)(i)	Award up to **two** marks for a linked description, such as: • In a wired network packets travel over (copper/fibre-optic) cables (1), whereas in a wireless network packets travel over the air (1)	Allow radio signals, infra-red, microwave for air (must be more than just waves) Allow twisted pair, CAT for copper. Allow electrical/light signals (not just electricity)	2

Question Number	Answer	Additional Guidance	Mark
4(c)(ii)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. A protocol (suite/group/family) (1) 2. Used on a wired network / wired connection (1) 3. Defines physical parts/type of cable(twisted pair, CAT6)/type of connector (1) 4. Defines how packets are checked for errors (1) 5. Defines the speed of transmission (1) 6. Operates at the link layer of the TCP/IP stack (1)		2

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 7 of 7 · Question 4 part a · 1 mark

2023 JUN · Paper 01 · extract 3 of 4 · find part a in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(d)	Award **three** marks for all of switch, router and modem in the correct order as in diagram Award **two** marks for all of switch, router and modem in any order Award **one** mark for any of switch or router or modem in any box 		3

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 7 of 7 · Question 4 part a · 1 mark

2023 JUN · Paper 01 · extract 4 of 4 · find part a in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(e)	Award up to **two** marks for a linked description, such as: Validation is used to check that the data/password entered by a user is fit for processing/meets specified requirements (1), whereas authentication is used to verify the identity of the user/prove that users are who they say they are (1)	For validation 'specified requirements' a good description of the check can be given e.g. presence check, length check, format check	2

Question Number	Answer	Additional Guidance	Mark